Development and Applications of Amino Acid Derived Organometallics

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Abstract: This Account describes the discovery and development of a family of organometallic reagents derived from naturally occurring α-amino acids that have found widespread use in the stereocontrolled synthesis of non-natural analogues of proteinogenic amino acids. The general approach that we have taken involves the conversion of the side chain of enantiomerically pure naturally occurring amino acids, such as serine, aspartic acid and glutamic acid, into the corresponding primary alkyl iodide, followed by conversion into the corresponding organozinc reagents. Subsequent palladium- or copper-catalysed reactions of these reagents allow the synthesis of a wide variety of functionalised α-, β- and γ-amino acid derivatives, without loss of stereochemical integrity. Insights gained from spectroscopic and kinetic studies into the stability of functionalised organozinc reagents are included.

Key words: amino acids, copper, cross-coupling, palladium, zinc

1 Introduction

The development of new methods for the stereocontrolled synthesis of nonproteinogenic amino acids continues to be an important area in modern organic chemistry, and a number of reviews exist on this subject.[1–5] One approach is to seek to convert existing amino acids into more complicated structures with retention of the original configuration. In this respect, the development of methods for the generation of zinc organometallics and their application to C–C bond forming reactions has opened up new possibilities in amino acid synthesis taking stereochemistry from the chiral pool. This account will focus upon our approach to the preparation of enantiomerically pure amino acids, in particular α-amino acids, through the use of functionalised organozinc reagents and will highlight some of the recent uses of this methodology. Earlier applications of our approach have been reviewed previously.[6]

2 Preparation and Applications of α-Amino Acid Derived Zinc Organometallics

2.1 Synthesis of Phenylalanine Derivatives

Many early approaches to the asymmetric synthesis of amino acids involved the generation of the stereogenic centre. Whilst this approach allows great flexibility, the control of stereochemistry generally requires the use of chiral auxiliaries and extra synthetic steps necessary for their addition/removal. At the time our group began its research effort into this field, procedures for the functionalisation of existing amino acids into more complicated structures were less well developed. The most common starting material in this approach was serine, and derivatisation to equivalents of both β-cation **1**[7,8] and β-anion **2**[9,10] were reported (Figure 1). During the late 1970s, Negishi observed that organozinc reagents underwent Ni(0)- or Pd(0)-catalysed reaction with aryl halides,[11] a development that marked the beginning of a greater synthetic use for these reagents. Encouraged by the stability of the Nakamura chiral zinc reagent **3** and its tolerance to ketone and ester functionalities,[12] we realised that an organometallic reagent derived from serine could function as a synthetic equivalent for the alanine β-anion via the Negishi methodology, provided the nitrogen protecting group could be tolerated.

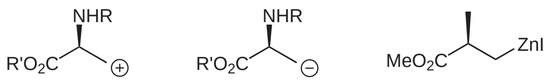

Figure 1

The precursor iodide **4** was easily prepared in four steps from commercially available L-serine. Luckily, concerns about the compatibility of the acidic proton of the NHBoc group were unfounded and in 1989 our group reported the synthesis of a range of phenylalanine derivatives **6** by the coupling of serine-derived zinc homoenolate **5a** with a range of *para*-substituted iodobenzenes under Pd(0) catalysis (ca. 35–65% yield, Scheme 1).[13] The reported yields are based on the amount of iodide **4**, which was used as the

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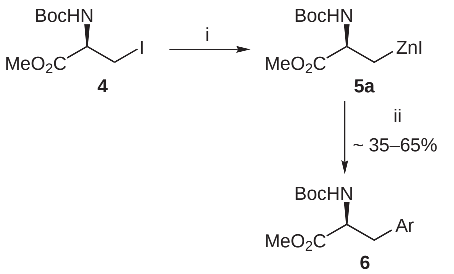


Scheme 1 *Reagents and conditions*: i) Zn/Cu couple, C_6H_6–DMA; ii) $PdCl_2[P(o\text{-tol})_3]_2$, ArI, 50 °C, 1 h.

limiting reagent. It was established that the reaction proceeded without loss of enantiomeric purity by standard methods.

Ultrasound facilitates the reaction of the alkyl iodide with Zn/Cu couple,[14–16] possibly due to stimulation of a single electron transfer from the metal.[17] There are a number of alternative ways to realise the transformation of alkyl iodide to organozinc halide, the most common being the reaction of the alkyl iodide with activated zinc (denoted as Zn*). This may be produced chemically by the reduction of zinc salts,[18] or by chemical or mechanical removal of oxides from the surface of the metal.[15,19,20]

$P(o\text{-tol})_3$ was found to be a much superior phosphine ligand to PPh_3, an observation supported by reports that more bulky phosphine ligands have been shown to facilitate the reductive elimination step in the catalytic cycle.[21] One of the main problems with this reaction concerned the coupling of some *ortho*-substituted aryl iodides, especially 2-fluoro iodobenzene. Since 1-iodo-2-methoxybenzene reacted satisfactorily, the explanation cannot be entirely a steric effect. Since *ortho* substituents of aryl halides have been shown to be able to coordinate to palladium upon oxidative addition,[22,23] a possible explanation lies with catalyst inactivation.

Subsequently, our group's research efforts were focused upon other applications of this chemistry and the original methodology remained unchanged until an improved method was identified in 1998. Modifications included using $Pd_2(dba)_3$ as the palladium source, DMF as the solvent and performing the reaction at room temperature,

Biographical Sketches

Ian Rilatt was awarded a BSc in Chemistry from the University of Nottingham in 1999. He obtained an MSc in Analytical Chemistry from Loughborough University in 2000, which was followed by a year in industry working as a scientist for Aphton Corporation. In 2001 he undertook a doctorate with Professor Richard Jackson in the area of amino acid derived organozinc reagents at the University of Sheffield. He currently holds a Marie Curie postdoctoral fellowship in the field of medicinal chemistry at the Centre de Recherche Pierre Fabre, Castres, France.

Lorenzo Caggiano obtained his BSc in Chemistry with Industrial Chemistry from the University of Liverpool in 1998. He then performed his PhD in new synthetic methodologies under the supervision of Dr. Stuart Warren at the University of Cambridge. As a European Network Fellow, he moved in 2002 to the group of Professor Cesare Gennari, at the University of Milan, Italy. There he was involved in the investigation and synthesis of simplified 'eleutherside' analogues, which still retain potent activities exhibited by the natural products, culminating in the formal synthesis of eleutherobin. Since February 2004 he has held the position of Research Officer in the group of Professor Richard Jackson at the University of Sheffield.

Professor **Richard Jackson** obtained his PhD in 1984 under the supervision of the late Professor Ralph Raphael, developing a new route to dihydrofuranones, and completing a synthetic approach to the pseudomonic acids. In 1984/5, he spent one year at the ETH, Zürich, working with Professor Dieter Seebach on the synthesis of a derivative of the macrodiolide elaiophylin. On returning to the UK in 1985, he took up a lectureship in Newcastle. Initial work was in the area nucleophilic epoxidation, and epoxide anion chemistry, but a programme in the area of unnatural amino acid synthesis was also initiated, and which is the subject of this account. He was promoted to a Chair in 1996, and in 2001 moved to a Chair in Synthetic Organic Chemistry at the University of Sheffield, where he has been Head of Department since 2003.

which together led to a moderate increase in overall yields for a range of electrophiles (ca. 40–70%), but also an improvement in reproducibility.[24]

Recent applications of our chemistry include the synthesis of the central core of 4′-phosphonomethylphenylalanine derivatives **7** and **8**[25,26] and a 4′-phosphonophenylalanine derivative **9**[27] as precursors to phosphotyrosine mimetics (Figure 2), by cross-coupling of serine-derived zinc reagents with the corresponding aryl iodides.

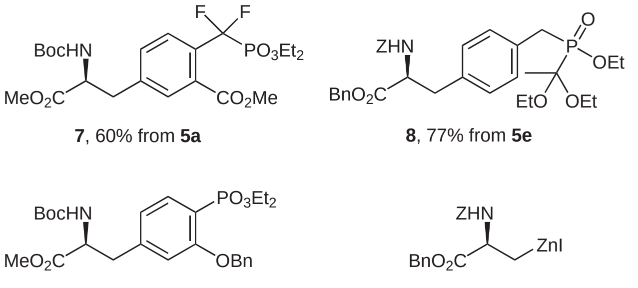
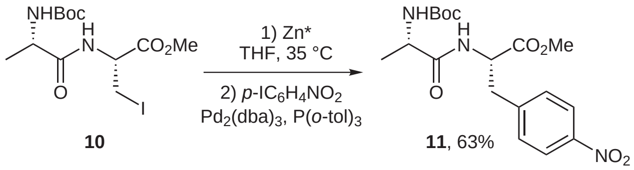


Figure 2 Isolated yields of products obtained by the cross-coupling of serine-derived organozinc reagents with aryl iodides.

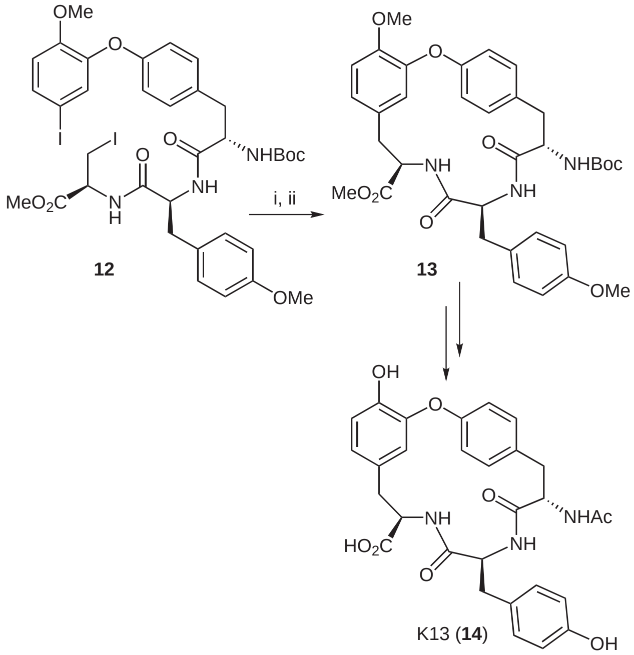
In an extension to the application of organozinc reagents derived from serine, we demonstrated that it was also possible to incorporate a carbon–zinc bond in the side chain of simple di- and tripeptides.[28,29] Thus, dipeptide **10** was converted into the corresponding zinc reagent and coupled under palladium catalysis with 4-iodonitrobenzene to give dipeptide **11** in good yield (63%, Scheme 2).



Scheme 2

Encouraged by this result we investigated the synthesis of K13 (**14**), an inhibitor of angiotensin converting enzyme (ACE). ACE inhibitors are an important class of blood pressure-lowering drugs. Our synthesis proceeded via a palladium-catalysed macrocyclisation of a linear tripeptide **12** containing an iodoaryl residue (Scheme 3).[30] The resulting cyclic tripeptide **13** was transformed into K13 (**14**) in good yield (74%). Although the key macrocyclisation step proceeded in modest yield, overall the synthesis is amongst the shortest and most efficient so far reported.

In 1999, the Gallagher group reported an application of our phenylalanine synthesis in their route to the glycoside **15**, in which a Negishi cross-coupling involving zinc reagent **5b** and the corresponding aryl iodide was employed (Figure 3).[31]



Scheme 3 *Reagents and conditions*: i) Zn*, DMF, r.t., 20 min; ii) $Pd_2(dba)_3$, P(*o*-tol)$_3$, 3×10^{-3} M in THF, 60 °C, 16 h, 35%.

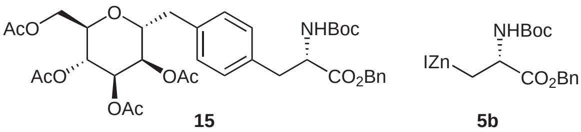
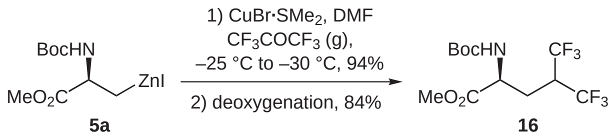


Figure 3 C-Linked glycoside synthesised by the Gallagher group.

A recently reported application of the zinc reagent **5a** was the synthesis of hexafluoroleucine **16** via a copper-promoted reaction with hexafluoroacetone,[32] followed by deoxygenation[33] of the intermediate alcohol (Scheme 4). It is interesting to note that although zinc reagents do not normally react directly with ketones, in this case the reaction proceeds due to the high electrophilicity of hexafluoroacetone. Interestingly, the reaction was found to proceed in the absence of the copper halide, albeit in lower yields. A more detailed discussion of zinc/copper reagents is included in section 6.

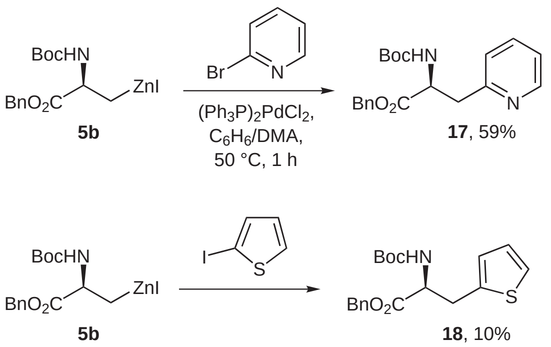


Scheme 4

2.2 Preparation of β-Heteroaryl Alanine Derivatives

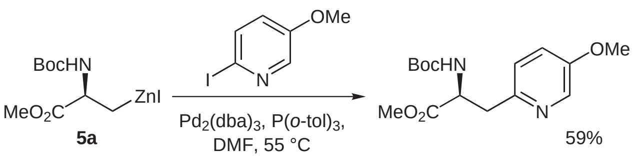
In our original publication[13] we had briefly explored the synthesis of a pyridylalanine derivative from serine by treatment of the organozinc reagent **5b** with 2-bromopyridine and $(Ph_3P)_2PdCl_2$ in benzene/DMA (Scheme 5).

The reaction proceeded satisfactorily to form the amino acid derivative **17** in good yield (59%). However, an attempt to synthesise protected β-(2-thienyl)alanine (**18**) via reaction with 2-iodothiophene proceeded poorly (10%), an observation which suggested that zinc reagent **5b** was relatively unreactive.



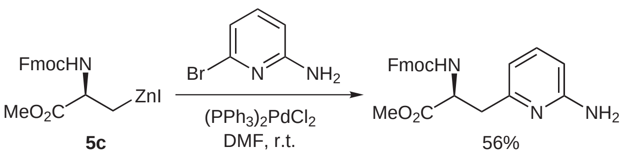
Scheme 5

This approach was later applied by the Westwell group in the synthesis of the precursor to the antitumour antibiotic L-azatyrosine via reaction of the zinc reagent **5a** with 2-iodo-5-methoxypyridine (Scheme 6).[34] Subsequent global deprotection (BBr_3 in CH_2Cl_2) yielded the desired product.



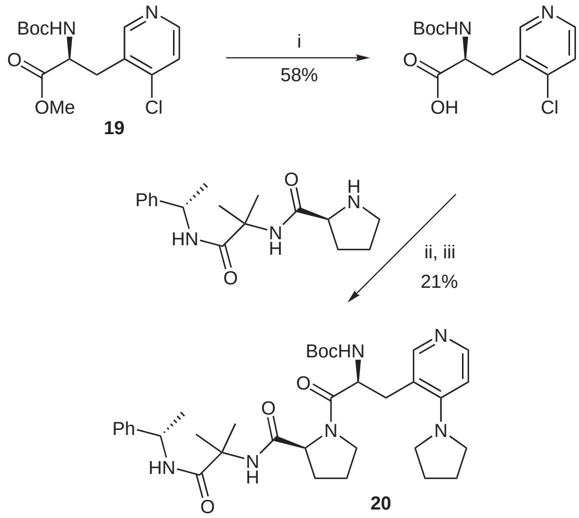
Scheme 6

More recently, our group has extended this chemistry to include the use of the Fmoc-protected zinc reagent **5c**[35] and more highly substituted bromopyridine electrophiles (Scheme 7).[36] The products were isolated in moderate, but consistent yields (ca. 40–60%). It is interesting to note that the presence of an unprotected aminopyridine, containing a relatively acidic proton, is tolerated in this reaction. This was an early indication that acidic protons could be tolerated.



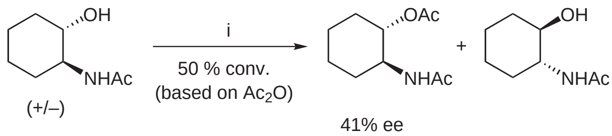
Scheme 7

Cross-coupling product **19**, prepared from the zinc reagent **5a** and 4-chloro-3-iodopyridine, was converted into nucleophilic catalyst **20** (Scheme 8), based on a tripeptide DMAP analogue. Catalyst **20** was used to effect the kinetic resolution of racemic *trans*-aminoacetyl-2-cyclo-



Scheme 8 *Reagents and conditions*: i) LiOH, THF–H_2O; ii) EDCI, HOBt, DMF, r.t.; iii) pyrrolidine, reflux, 3 d.

hexanol with acetic anhydride, although only a modest enantioselectivity (41% ee) was obtained (Scheme 9).[36] Analogous tripeptidic compounds were extensively developed by Miller, and had been applied to effect the same transformation (with much greater success!).[37–39]



Scheme 9 *Reagents and conditions*: i) **20** (0.5 mol%), Ac_2O (0.1 equiv), 0 °C.

Another recent application of our chemistry to the synthesis of heteroaryl amino acid derivatives is the preparation of protected (purin-6-yl)alanines **21** and **22** via cross-coupling with 6-iodopurines (Figure 4). Several examples of this class of compounds were reported to possess cytostatic, antimicrobial, and A_2-receptor agonist activity.[40] In cases such as this, where the electrophilic partner is more valuable, very high yields (based on the electrophile) may be obtained by simply using an excess of the organozinc reagent.

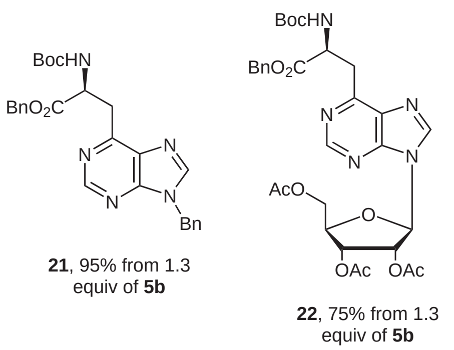
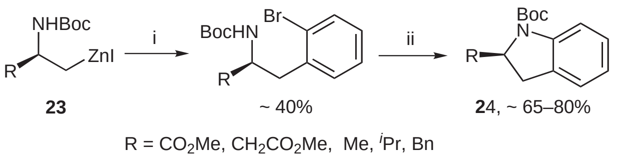


Figure 4 Isolated yields of products obtained by the cross-coupling of serine-derived organozinc reagents with aryl iodides.

Following Buchwald's development of methodology for the palladium-catalysed amination of aromatic halides,[41] we envisioned that a combination of this methodology with our cross-coupling reaction would constitute a simple, yet effective, ring annelation process. This would allow access to a range of substituted indolines, which are compounds commonly found as constituents of biologically active molecules and natural products.[42] Palladium(0)-catalysed cross-coupling of chiral organozinc reagents **23** with 2-bromoiodobenzene at room temperature furnished the cross-coupled intermediates in moderate yields after purification, and which were subsequently cyclised according to the procedure described by Buchwald to yield a range of 2-substituted indolines **24** (Scheme 10).[43]



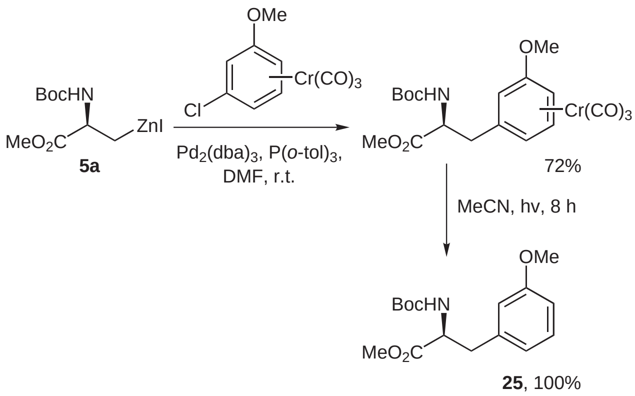
Scheme 10 *Reagents and conditions*: i) 2-bromoiodobenzene, $Pd_2(dba)_3$, $P(o\text{-tol})_3$, DMF, r.t.; ii) $Pd_2(dba)_3$, $P(o\text{-tol})_3$, Cs_2CO_3, toluene, 100 °C, 15 h.

3 Cross-Couplings with Aryl Triflates, Bromides and Chlorides

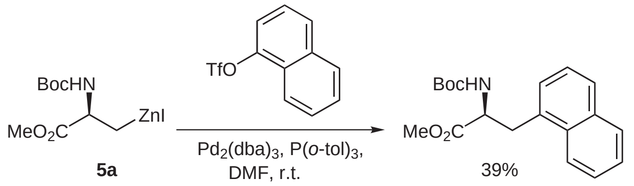
A recent publication by Fu showed that $Pd[P(t\text{-}Bu)_3]_2$ can effect the Negishi cross-coupling reaction of a range of aryl and vinyl chlorides with aryl- and alkylzinc reagents, although reaction temperatures of 100 °C are required.[44] Nevertheless, the use of aromatic bromides and chlorides as coupling partners in these processes remains attractive due to their lower cost and wider diversity.

The complexation of aromatic chlorides with the electron-withdrawing tricarbonylchromium fragment renders the substrate more susceptible to oxidative addition by palladium, and hence to cross-coupling reactions conducted under milder conditions. A short investigation into the application of this chemistry revealed that these complexes were indeed reactive partners in cross-coupling reactions. The reaction with (3-chloromethoxybenzene)tricarbonylchromium proceeded smoothly, and the product could be decomplexed by irradiation with a tungsten lamp to furnish the protected amino acid **25** in excellent yield (Scheme 11). A small range of other substrates was also investigated, although lower yields were generally obtained.[45]

In 1986, Yoshida showed that vinyl triflates were reactive cross-coupling partners,[46] and a 1999 report by Lipshutz[47] disclosed the nickel-catalysed cross-coupling between an organozinc reagent and naphthyl triflate. Since aryl triflates are generally accessed more easily than aryl iodides, our group explored the application of this reaction and demonstrated that amino acid-derived organozinc



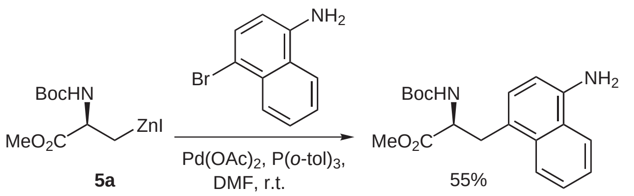
Scheme 11



Scheme 12

reagents may be coupled with these compounds, although in low yield (Scheme 12).[48]

A single example of an aryl bromide coupling reaction, namely the reaction of zinc homoenolate **5a** with 1-amino-4-bromonaphthalene, was recently described in the synthesis of a potent tyrosine phosphatase inhibitor (Scheme 13).[49] Work is ongoing in our group on extending the cross-coupling reactions of organozinc reagents with aryl bromides.



Scheme 13

4 Cross-Couplings with Vinyl Triflates and Iodides

Vinyl triflates were successfully used as coupling partners with serine-derived organozinc reagents to generate tetra-substituted pyrans **26a**–**c**, advanced intermediates used in the synthesis of (–)-dysiherbaine (via **26a**[50] or **26b**[51]) and neodysiherbaine A (a biologically active amino acid with excitatory properties isolated from a marine sponge) via **26c**.[52] The pyridine-based phosphotyrosine mimetic **27** was also synthesised from a cross-coupling reaction between the corresponding vinyl triflate and the organozinc reagent **5b** (Figure 5).[53]

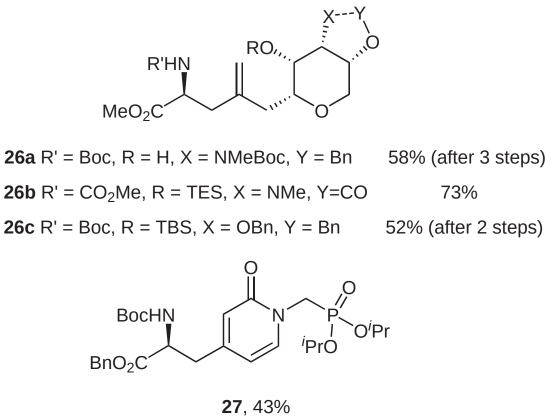
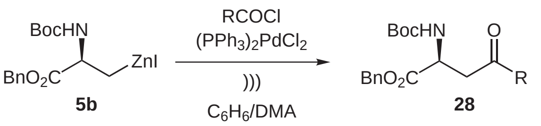


Figure 5 Isolated yields of products obtained by the cross-coupling of serine-derived organozinc reagents with vinyl triflates.

We have also reported the use of solid-supported 3-iodoacrylates as cross-coupling partners.[54]

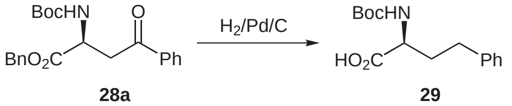
5 Preparation of Keto Amino Acids

The synthesis of ketones by nucleophilic addition of organometallic reagents to carboxylic acids and their derivatives is an important transformation in organic synthesis. Encouraged by Yoshida's 1985 publication,[55] in which an organozinc halide was acylated by an acid chloride under Pd-catalysis, it seemed reasonable to explore the possibility that a more highly functionalised zinc reagent might be used in the cross-coupling reaction. Indeed, sonication of **5b** in benzene–DMA with a range of acid chlorides under palladium catalysis resulted in efficient reactions in just 30 minutes, furnishing the γ-keto α-amino acids **28** in good yields (ca. 70–90%, Scheme 14).[56,57] Subsequent control experiments confirmed that sonication was not needed for the cross-coupling reaction.



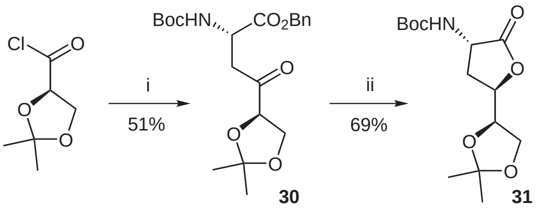
Scheme 14

We also found that this chemistry is applicable to the synthesis of protected homophenylalanine, as hydrogenation of the ketone **28a** (R = Ph), resulting from the reaction of the zinc reagent with benzoyl chloride, furnished optically pure Boc-homophenylalanine (**29**) in quantitative yield (Scheme 15).[58]



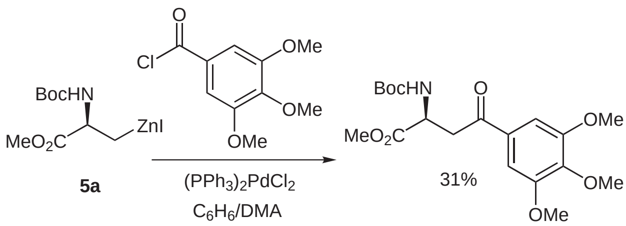
Scheme 15

The utility of these reactions was exemplified by the short, stereoselective synthesis of protected 4,5,6-trihydroxylated norleucines via the protected 4-keto α-amino acid **30**, which was reduced to the corresponding lactone **31** (Scheme 16).[59,60] The reaction was also performed with organozinc reagent *ent*-**5b**, prepared from D-serine, to give the C2 epimer of the intermediate **30** (in 45% yield), which offers a new synthetic approach to the synthesis of (+)-bulgecinine, the enantiomer of a component of the naturally occurring bulgecin glycopeptide.



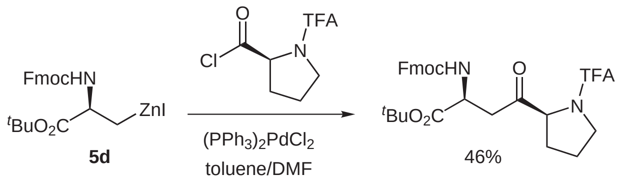
Scheme 16 *Reagents and conditions*: i) **5b**, $(PPh_3)_2PdCl_2$, benzene–DMA; ii) L-Selectride, THF, –78 °C, 3 h.

A recent paper reports the synthesis of a fragment of colchicine by the acylation of zinc reagent **5a** (Scheme 17).[61] The low yield in this reaction is a reflection of the relatively unreactive electrophile, and higher yields were obtained in a model reaction with benzoyl chloride (52–65%).



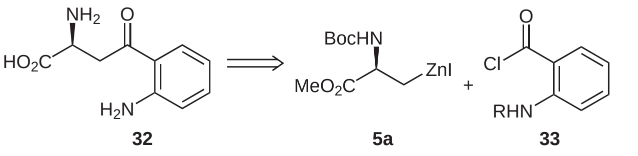
Scheme 17

To illustrate further the functional group tolerance of this process, a functionalised 4-oxo amino acid was synthesised by the acylation of the serine-derived reagent **5d** with a proline-derived acid chloride (Scheme 18).[35,62]



Scheme 18

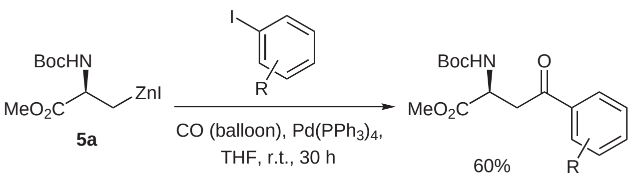
Notwithstanding our synthesis of a range of 4-keto amino acids, the direct synthesis of L-kynurenine (**32**), an intermediate in the metabolism of L-tryptophan to nicotinic acid ribonucleotide, presented us with a challenge. The most direct approach to its synthesis would require the coupling of an anthranilic acid derivative with the serine reagent **5a** (Scheme 19).



Scheme 19

However the most serious problem with this approach is the choice of nitrogen protecting group for the anthranilic acid chloride (**33**). Many nitrogen protecting groups are incompatible with acid chlorides, enhance the acidity of protons on the nitrogen or are sufficiently bulky that they severely retard reaction rates. Given these challenges, our attention turned towards an alternative strategy, involving the palladium-catalysed carbonylative cross-coupling of organozinc reagents with aromatic iodides. Literature precedent suggested that this strategy would be feasible, although at that time little work had focused on the use of zinc reagents.[63,64] Additionally, THF had been shown to be a suitable reaction medium. This solvent is generally inefficient for acylation using acid chlorides due to cleavage of the solvent by the electrophile to form 4-chlorobutanol derived esters, a process mediated by zinc salts.[65]

We were pleased to discover that our initial experiment involving the reaction of iodobenzene with serine-derived zinc reagent **5a** under an atmosphere of carbon monoxide (balloon) at room temperature in the presence of $Pd(PPh_3)_4$ gave the protected 4-keto amino acid in 60% yield (Scheme 20).[66]

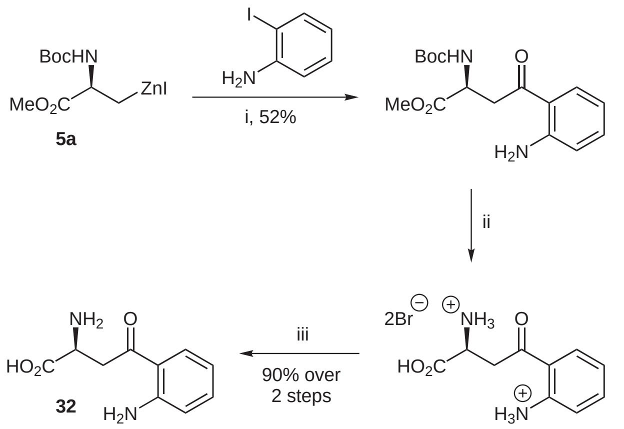


Scheme 20

The scope and limitations were investigated by the reaction of a variety of functionalised aryl iodides giving the products in poor to fair yield (ca. 10–60%) and even unprotected *ortho*- and *para*-iodoanilines were found to be tolerated. This allowed easy access to our initial target by the coupling of reagent **5a** with 2-iodoaniline which proceeded in acceptable yield (52%), followed by subsequent deprotection (Scheme 21).

In some cases, for example with 3-iodoaniline, the yield of the carbonylated cross-coupled product was much lower (17%). In this case, and in others where low yields of the desired product were isolated, significant amounts of the carbonylative homocoupled product, protected 4-keto-2,6-diaminopimelic acid **34**, were found (Figure 6).[66]

Further experimentation with this reaction revealed that high yields of the carbonylative homocoupled products



Scheme 21 *Reagents and conditions*: i) CO (balloon), $Pd(PPh_3)_4$, THF, r.t., 30 h; ii) HBr (30% in AcOH), r.t., 20 min; iii) propylene oxide, *i*-PrOH.

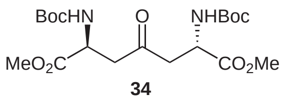
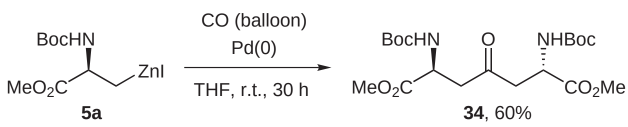


Figure 6

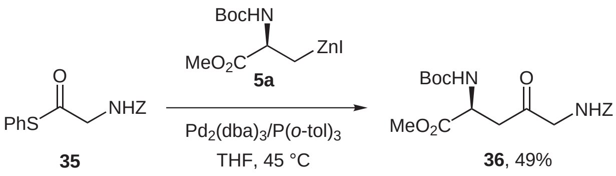


Scheme 22

from organozinc reagents could be obtained by the deliberate exclusion of an electrophile (Scheme 22).[66]

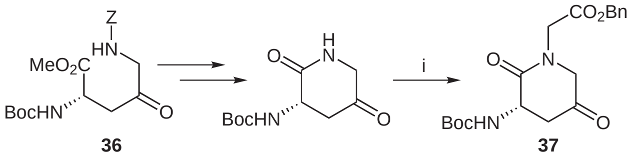
This process is catalytic in palladium but the source of oxidant required for the regeneration of the palladium(0) catalyst was initially rather mysterious. After performing a series of experiments it was concluded that the most likely oxidant was molecular oxygen, which had diffused into the system through the balloon. When oxygen was rigorously excluded from the system, none of the symmetrical ketone was formed. Repetition of the carbonylative coupling between the zinc reagent **5a** and 3-iodoaniline using these optimised conditions led to a much improved yield of the isolated product (from 17% to 52%).[66]

In 1998, Fukuyama reported that organozinc reagents are effectively acylated under Pd-catalysis by thiolesters in THF.[67] A number of transition-metal-catalysed processes, including the palladium-catalysed reaction of thiolesters with boronic acids,[68,69] organostannanes,[70] and alkynes[71] have been reported more recently. Given that this reaction was a clear alternative to the carbonylative couplings, our group decided to investigate the applicability to our zinc reagents. An initial experiment involving the reaction between organozinc reagent **5a** and *Z*-glycine-derived thiolester **35** using $(PPh_3)_2PdCl_2$ as the catalyst resulted in a modest yield of the expected ketone **36** (39%).[72] Replacement of the catalyst with $Pd_2(dba)_3/P(o\text{-tol})_3$ led to isolation of the product in superior yield (49%, Scheme 23).



Scheme 23

Cyclisation of ketone **36** allowed the synthesis of the protected Ala-Gly surrogate **37** (Scheme 24).[72]



Scheme 24 *Reagents and conditions*: i) LHMDS, benzyl bromoacetate, THF, –78 °C to r.t., 16 h.

Compounds possessing a 3-amino-2-piperidone nucleus often exhibit biological activity, for example, compound **38** has been reported as a serine protease inhibitor[73] and compound **39** is an ACE inhibitor (Figure 7).[74]

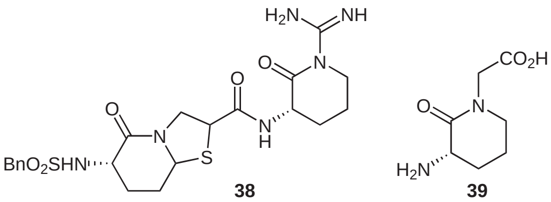


Figure 7 Biologically active compounds containing a 3-amino-2-piperidone nucleus.

6 Preparation and Applications of α-Amino Acid Derived Zinc/Copper Organometallics

The Michael addition is a highly powerful synthetic transformation. A few years before our amino acid research program began, the Nakamura group reported the copper iodide catalysed conjugate addition of zinc homoenolates to enones in high yield, as well as two examples of reactions with acid chlorides to form ketones.[75] Interestingly, they reported that TMSCl was necessary for the reactions to proceed. In 1988, Knochel disclosed the preparation of a new class of mixed copper/zinc reagents by the transmetallation of zinc organometallics with the THF-soluble salt, CuCN·2LiX.[76] The resulting functionalised zinc/copper reagents, formulated as RCu(CN)ZnI, react directly with acid chlorides and enones to afford ketones and 1,4-addition products, respectively. A 1989 publication by the Yoshida group extended Nakamura's 1,4-addition methodology to organozinc reagents bearing electrophilic functionalities, such as esters, nitriles and two interesting reactions involving an α-amino acid derivative **40** (Figure 8).[77] These results suggested that a more reactive a-amino acid nucleophile **41a** would be available by transmetallation from zinc reagent **5a**, and in 1992 we reported the S_N2' reaction of a range of allylic halides and tosylates with our zinc/copper reagent **41b** in moderate isolated yield (Scheme 25).[78]

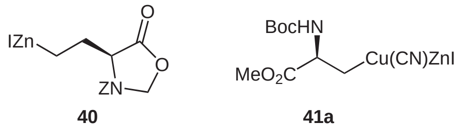
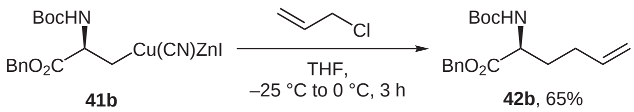
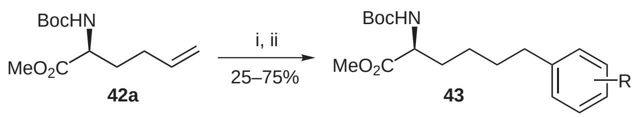


Figure 8



Scheme 25

Using this synthetic route, we prepared enantiomerically pure protected but-3-enylglycine (**42a**), which our group subsequently used in the synthesis of a range of tris-homophenylalanines **43** via hydroboration–Suzuki cross-coupling methodology, in an extension of chemistry developed by Taylor[79–81] and Johnson[82] (Scheme 26).[83]

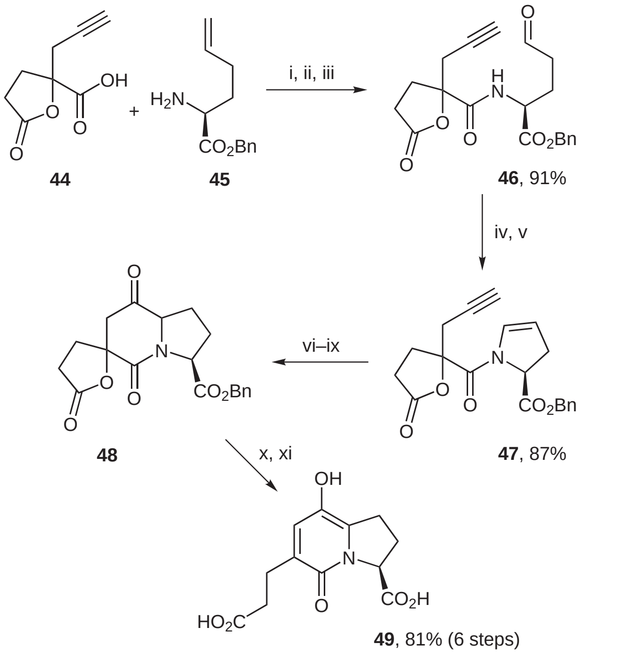


Scheme 26 *Reagents and conditions*: i) 9-BBN (2 equiv), THF; ii) ArX, $PdCl_2$(dppf) (5 mol%), aq K_3PO_4, THF, r.t., 16 h.

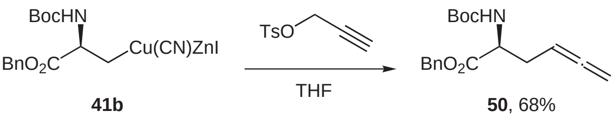
Protected but-3-enylglycine, prepared by this method, has been used by the Clive group in the synthesis of a powerful ACE inhibitor (Scheme 27).[84] Thus, ozonolysis of the intermediate resulting from the coupling of the carboxylic acid **44** with protected but-3-enylglycine (**45**) yielded peptide **46**. Cyclisation to a hemiaminal followed by dehydration furnished the tertiary amide **47**. Target compound **49** was formed following a radical cyclisation and oxidation to ketone **48**, followed by deprotection.

In addition to allylic electrophiles, treatment of the zinc/copper reagent **41b** with propargylic halides and tosylates furnished allenic amino acids in good yields without any trace of the corresponding acetylenes (ca. 50–80%).[85] For example, the preparation of an allenic amino acid **50** is realised by treatment of **41b** with propargyl tosylate (Scheme 28). Many olefinic amino acids have been isolated from natural sources, especially fungi, and some possess biological activity.

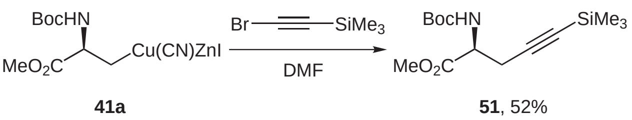
Zinc/copper reagents also react with haloalkynes, in a mechanism believed to be an addition–elimination reaction involving a *syn* carbocupration followed by an *anti* elimination.[86] We reported the successful application of



Scheme 27 *Reagents and conditions*: i) EDCI·HCl, HOBt, CH_2Cl_2, DMF; ii) O_3, CH_2Cl_2, –78 °C; iii) Ph_3P, –78 °C to r.t.; iv) BaO, CH_2Cl_2; v) P_2O_5; vi) Bu_3SnH, AIBN, PhMe, reflux; vii) CF_3CO_2H, THF; viii) O_3, CH_2Cl_2, –78 °C; ix) Ph_3P, –78 °C to r.t.; x) Et_3N, THF, 60 °C; xi) H_2, Pd/C, MeOH.



Scheme 28

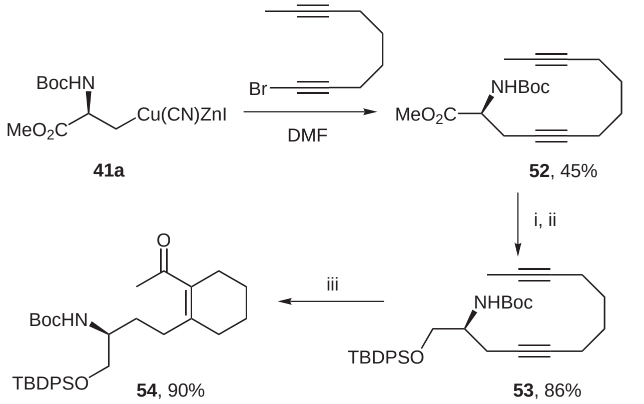


Scheme 29

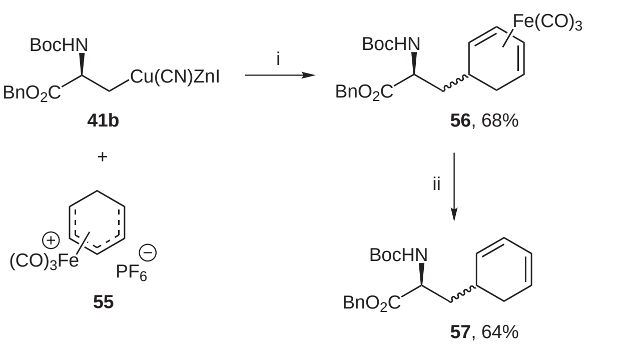
this process to the zinc/copper reagent **41b** in a reaction with ethyl bromopropiolate.[85] This method was recently applied to the synthesis of the acetylene-containing amino acid **51** (Scheme 29).[87]

Copper/zinc-alkyne chemistry involving **41a** has also recently been applied by the Trost group in the synthesis of cylindricine.[88] Transformation of product **52** into **53** was achieved via a ruthenium-catalysed hydrative diyne cyclisation to give a late stage intermediate **54** (Scheme 30).[89]

Following literature reports that less highly functionalised zinc/copper reagents reacted with tricarbonyl(η^5-dienyl)iron cations,[90] we decided to investigate the application of the serine-derived zinc/copper reagent **41b**. Initial reaction with a variety of these electrophiles led to modest yields (ca. 30–40%), which we ascribed to an inhomogeneity of the reaction mixture, and a mixed solvent system consisting of benzene–DMA–THF–MeCN was found to perform better (ca. 30–70%).[91] For example, treatment of



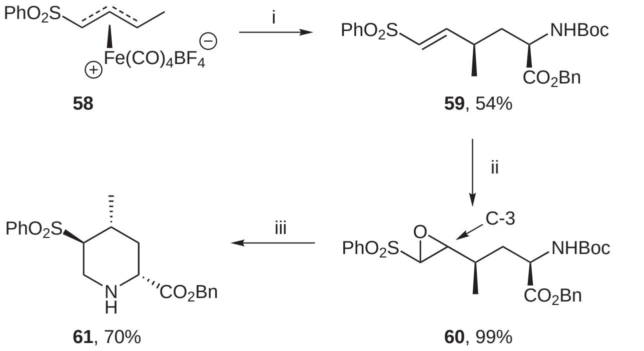
Scheme 30 *Reagents and conditions*: i) $LiBH_4$; ii) TBDPSCl, imidazole, DMF; iii) 5% $[CpRu(CH_3CN)_3]PF_6$, 10% H_2O–acetone, 60 °C, 2 h.



Scheme 31 *Reagents and conditions*: i) benzene–DMA–THF–MeCN, –20 °C, 16 h; ii) $Me_3N^+O^-$–DMA, 55 °C, 30 min.

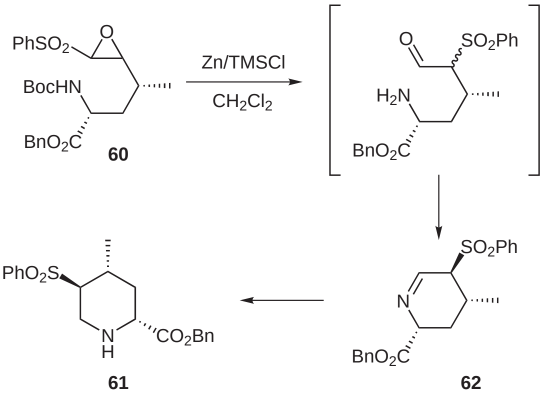
41b with iron complex **55** at room temperature generated cyclohexadiene adduct **56** as a 1:1 diastereoisomeric mixture in 68% yield. The iron fragment could be removed under oxidative conditions to yield the protected amino acid **57** (Scheme 31).

Whilst the stereochemical integrity of the α-centre is preserved in this process, reaction with these prochiral electrophiles always resulted in a mixture of diastereoisomers, generally in a ratio of close to 1:1. We were therefore interested in the outcome of the reaction between our zinc/copper reagent and stereochemically defined (η^3-allyl) cation equivalents, especially following a report by Enders on the addition of other zinc/copper reagents to such electrophiles.[92] Addition of 1.5 equivalents of *ent*-**41b** to the enantiomerically pure iron salt **58** at –78 °C and gradual warming to –20 °C, followed by oxidative removal of the iron tetracarbonyl unit with ceric ammonium nitrate, gave the diastereoisomerically pure α-amino acid **59** in 54% yield (Scheme 32).[93] With this product in hand we explored the potential application to the synthesis of cyclic amino acid derivatives. Epoxidation of **59** followed by treatment of the resulting epoxide **60** with Zn/TMSCl in CH_2Cl_2 unexpectedly gave a 5-phenylsulfonyl-4-methyl pipecolic acid derivative **61**, rather than the expected pyrrolidine (which would arise from nucleophilic attack at C-3 of the oxirane). In addition, compound **61** was isolated as a single diastereoisomer. We believe that this product arises from the isomerisation of phenylsulfonyl-oxirane **60** to the corresponding α-phenylsulfonyl aldehyde,[94] followed by a zinc-mediated intramolecular reductive amination (Scheme 33).



Scheme 32 *Reagents and conditions*: i) *ent*-**41b**, MeCN, –78 °C to –20 °C, 16 h; ii) LiOO*t*-Bu, THF, –20 °C, 4 h; iii) Zn, TMSCl, CH_2Cl_2.

The stereochemical outcome is most likely due to epimerisation at the carbon atom bearing the phenylsulfonyl group to the all-equatorial isomer in the presumed precursor imine **62** (Scheme 33).



Scheme 33

The synthesis of homologues of amino acids is important in the context of protein engineering and structure–activity relationships. Following the reaction of our zinc reagent with allylic and propargylic electrophiles, we considered whether this approach might be applicable to the synthesis of valine, leucine and isoleucine homologues, which would require reaction with allylic electrophiles such as **63–65** (Figure 9). These electrophiles are more highly substituted than those we had previously used.

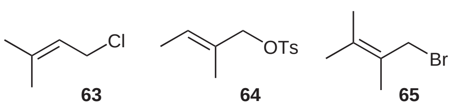
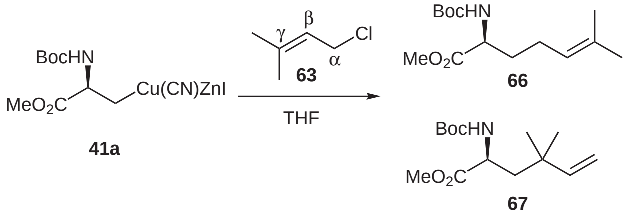


Figure 9

In 2000, we reported that the reaction of **41a** with 3,3-dimethylallyl chloride (**63**) gave a mixture of isomers **66** and **67** arising from α- and γ-substitution pathways in a 58:42 ratio (Scheme 34, 93%).[95]



Scheme 34

Due to the toxicity of CuCN·2LiCl, used in a stoichiometric quantity in our original publication,[85] we decided to investigate the use of catalytic CuBr·DMS, which had been reported to catalyse the reaction between β-amino zinc reagents and propargylic and allenic halides.[96] The modification was found not to have any detrimental effect upon the course of the reaction and the new procedure was used to synthesise the desired range of *N*-Boc- and *N*-Fmoc-protected amino acids **68**, **69**, **72**, **73**, **76**, **77**, following hydrogenation of the unsaturated intermediates (Figure 10).[95]

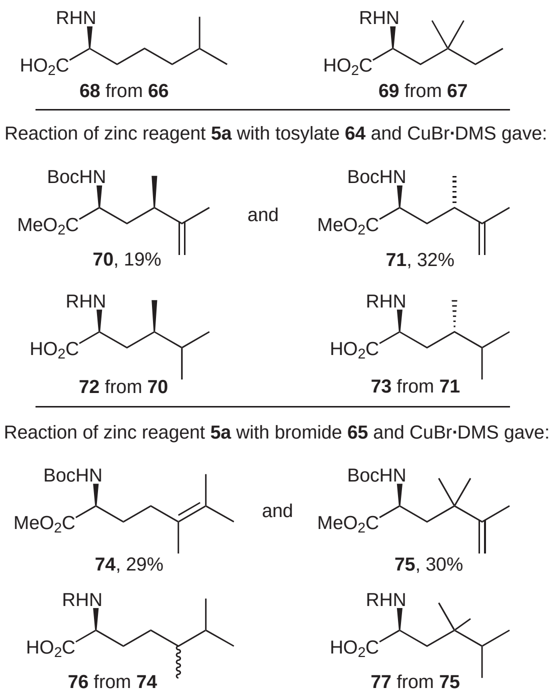


Figure 10

Further investigation with these reaction conditions revealed that superior yields were consistently obtained using catalytic CuBr·DMS in place of stoichiometric CuCN·2LiCl in allylations of the cuprates derived from organozinc iodides **78** and **79** (Figure 11).[97] One possible explanation could lie with the reduced stability of the organometallic intermediates in the presence of halide ions, for reasons discussed later, or that the zinc/copper reagent may simply be intrinsically less stable.

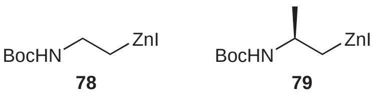
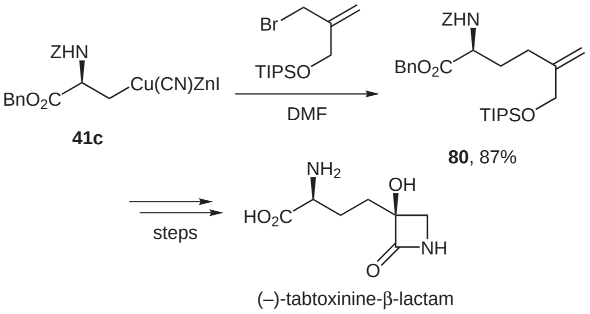


Figure 11

Recent literature reports utilising this chemistry include the synthesis of tabtoxinine-β-lactam, the phytopathogenic compound excreted by bacteria in Tobacco Wildfire disease. Following reaction of an allylic bromide with copper reagent **41c**, intermediate compound **80** was transformed into the target molecule following a Sharpless asymmetric dihydroxylation and further synthetic manipulations (Scheme 35).[98]

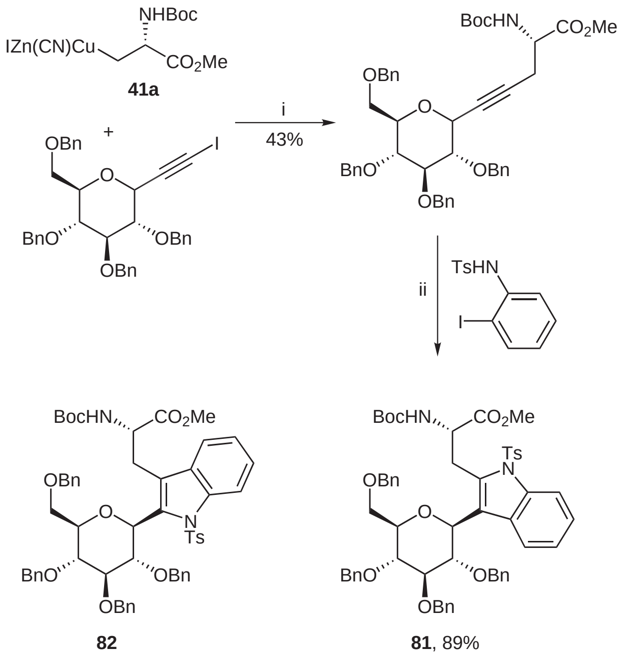


Scheme 35

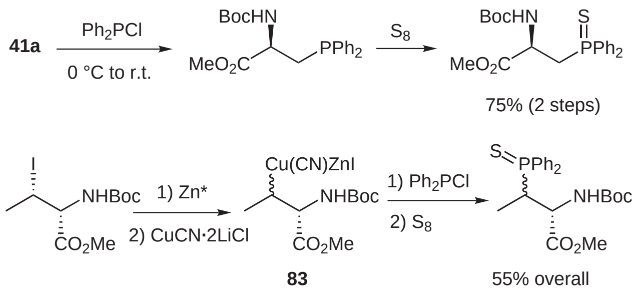
More recently, **41a** was used by the Isobe group in an approach to the synthesis of the *C*-glycosyl amino acid **82** (Scheme 36).[99] The regioselectivity of the ring annelation was described by Larock to be controlled by steric balance between the two acetylene substituents to afford a product having the sterically hindered substituent at the 2-position of the indole.[100,101] Interestingly, the ring annelation in this synthesis proceeded with the reverse regioselectivity, furnishing *iso*-tryptophan (**81**) rather than the expected tryptophan (**82**).

In addition, the serine-derived cuprate **41a** has been applied to the synthesis of phosphine-containing amino acids, which were protected as phosphine sulfides.[102] The corresponding threonine-derived cuprate **83**, formed as a 1:1 mixture of diastereoisomers was also employed (Scheme 37).[102] We have also investigated the reaction of cuprate **83** with simple allylic halides.[103]

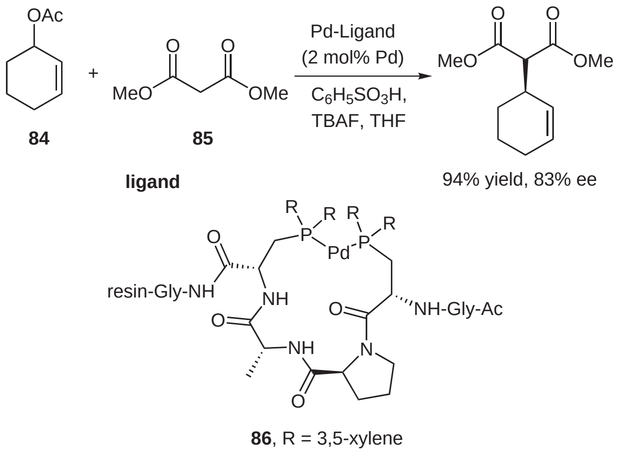
Such phosphines have potential as ligands in transition-metal-catalysed reactions. For example, the Gilbertson group has carried out the synthesis of amino acid derived chiral palladium–phosphine complexes such as **86**. The effectiveness of these catalysts in Pd-catalysed allylic substitution of cyclohex-2-enyl acetate (**84**) by dimethyl malonate (**85**) was assessed and the reaction proceeded in excellent yield with good enantioselectivity (Scheme 38).[104]



Scheme 36 *Reagents and conditions*: i) THF, –78 °C to 0 °C; ii) $Pd(OAc)_2$, PPh_3, n-Bu_4NCl, Na_2CO_3, DMF, 100 °C, 24 h.



Scheme 37



Scheme 38

7 Preparation of Homologous Amino Acid Derived Organometallics

Following our reports of the zinc reagent **5** and zinc/copper reagent **41**, we became interested in the homologous reagents **89**, **90** and **91** (Figure 12).

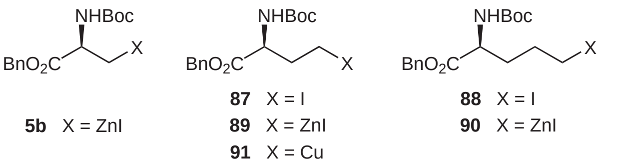


Figure 12

These reagents, which should function as synthetic equivalents of the α-amino acid γ- and δ-anions **92** and **93**, respectively (Figure 13), were synthesised from aspartic or glutamic acid using straightforward transformations.

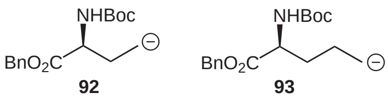
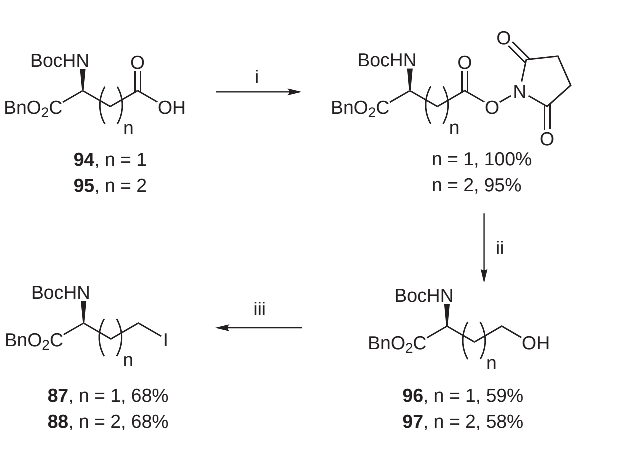


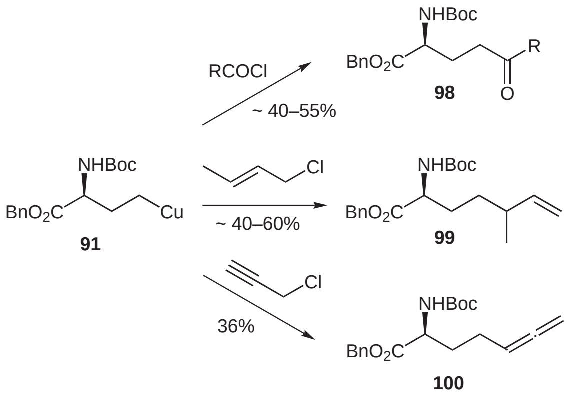
Figure 13

Thus, reduction of commercially available protected aspartic or glutamic acids **94** and **95** via the succinimide ester gave alcohols **96** and **97**. These were iodinated using I_2/PPh_3/imidazole to furnish iodides **87** and **88** in good yield (Scheme 39).[105] The iodide precursor to zinc reagent **40** had previously been synthesised in our group from protected glutamic acid using a Barton decarboxylative iodination.[106] Whilst this synthesis was convenient on a small scale, the tedious chromatographic removal of excess iodoform rendered this process inefficient for larger quantities and the alternative method was found to be more convenient.



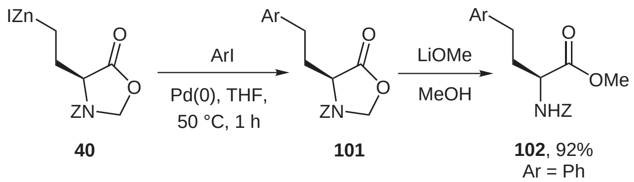
Scheme 39 *Reagents and conditions*: i) NHS, DCC, EtOAc; ii) $NaBH_4$, THF, H_2O; iii) I_2, PPh_3, imidazole, CH_2Cl_2.

Unfortunately, the first sign of stability problems with the homologues arose when an attempt was made to form zinc reagent **89**, which resulted in the production of significant amounts of protected 2-aminobutanoic acid via protonation of the carbon–zinc bond. We reasoned that the molecule was self-protonating, and alkyl copper species **91** prepared directly by the reaction of the iodide with Rieke copper was found to offer a better compromise between productive reactivity and instability. This allowed the synthesis of 5-keto- (**98**), allylic- (**99**), and allenic (**100**) amino acids, albeit in moderate yields (Scheme 40).[107]



Scheme 40

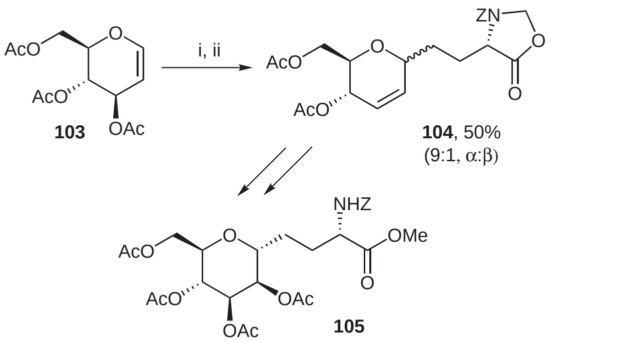
The Yoshida zinc reagent **40**,[77] reported in 1989 and which had not been used in the context of homophenylalanine synthesis, appeared to be amenable to this application since it did not contain an acidic carbamate proton. Preparation of the zinc reagent following Knochel's procedure[76,108] proceeded smoothly, and in 1994 we reported the synthesis of a range of homophenylalanines **101** (Ar = Ph) by the cross-coupling with a variety of iodobenzenes under palladium catalysis in THF (ca. 40–60%, Scheme 41).[109] The protecting group could be removed by treatment of **101** with lithium methoxide in methanol to give the protected homophenylalanine derivative **102** in excellent yield (92%).



Scheme 41

The next year, following a short screening to find a suitable solvent system, we reported the preparation of a range of 5-keto amino acids by the reaction of zinc reagent **40** with acid chlorides in DME.[110] The isolated yields (ca. 50–80%) were found to be much superior to those obtained from copper reagent **91**. Further refinement of this reaction, involving the use of benzene–DMA or toluene–DMA as the solvent led to a further increase in average yield, especially with more highly functionalised acid chlorides.[106] It was starting to become apparent that the proper choice of solvent was critical for the success of these reactions.

We also investigated the application of the organozinc reagent **40** in the synthesis of C-linked glycosyl amino acids **105** (Scheme 42).[111] Selective attack at the C-1 position of tri-*O*-acetyl-D-glucal (**103**) was achieved with a Lewis acid, as reported by Gallagher,[112] to give the C-linked adduct **104** in 50% yield.



Scheme 42 *Reagents and conditions*: i) **40**; ii) $BF_3{\cdot}OEt_2$, 0 °C to 30 °C.

Although cyclic derivative **40** was a satisfactory α-amino acid α-anion equivalent, acyclic derivatives **89** and **90** are slightly more flexible due to the higher versatility with respect to post-reaction protecting group manipulations (Figure 14).

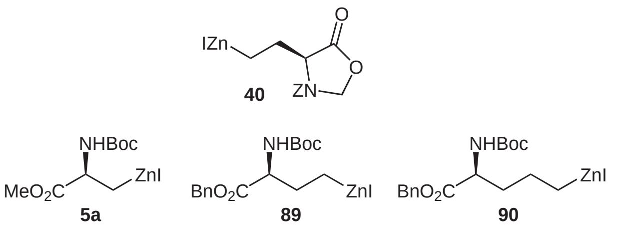
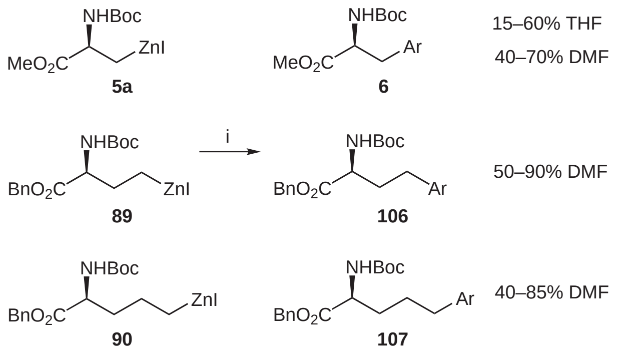


Figure 14

However, initial experiments with reagent **89** had highlighted its instability. We had already shown a superior performance of Yoshida's reagent **40** in changing from ethereal to dipolar aprotic solvents, consistent with Knochel's observation that simple N-protected iodoethylamine derivatives could be converted into stable zinc reagents in THF–DMSO mixtures.[113] Additionally, DMF was reported to allow the formation of arylzinc iodides,[114] a process which is inefficient in THF except with Rieke zinc, and DMF had also been shown to be beneficial for palladium-catalysed cross coupling reactions.[115] We therefore decided to investigate systematically what progress could be made with the homo- and bishomoalanine zinc reagents using DMF as the solvent.

Initial investigations into the preparation of the three homologous reagents **5a**, **89** and **90** in THF immediately showed the preparation of **5a** to be faster, with the homologues requiring gentle warming to 35 °C to promote zinc insertion in a comparable time to **5a** at ambient temperature.[105] The organozinc reagents were prepared in deuterated solvent and transferred under an inert atmosphere to an NMR tube fitted with a Young's tap. Although zinc reagent **5a** formed cleanly, significant amounts of protonated material were found with the homologues.

In changing the solvent to DMF, all three zinc reagents could be formed cleanly under identical conditions to those used in THF. Cross-couplings with reagent **5a** demonstrated that the coupling process is more effective in this solvent, the most significant results being the increased yields with *ortho*-substituted aryl iodides when using DMF. The most striking result is the successful preparation of the 2-fluorophenyl derivative, which completely eluded us using our previously developed protocol. Homophenylalanine **106** and bishomophenylalanines **107** derivatives were also prepared in satisfactory to excellent yields (Scheme 43).[105]



Scheme 43 *Reagents and conditions*: i) ArI, Pd(0), r.t., 3 h.

The ability to conduct the coupling reaction in DMF at room temperature makes the whole process extremely convenient to carry out, and these conditions, published in 1998, are the best that we have identified to date for the reaction.[105]

8 Stability and Reactivity Studies of β- and γ-Amidozinc Reagents

At the same time as our development of the α-amino acid synthons, we became interested in the potential of the related reagents **110** and **111** for the synthesis of enantiomerically pure β- and γ-amino acids (Figure 15).

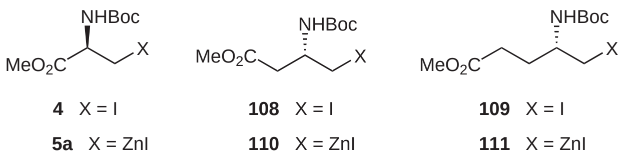


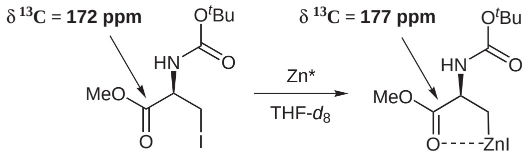
Figure 15

The precursor iodides were synthesised in good yield via a similar reduction–iodination protocol to the aspartic and glutamic α-amino acid analogues **87** and **88**.[116] However, the initial preparation and cross-coupling of the aspartic acid-derived reagent **110** in THF led to a disappointing yield of the desired homophenylalanines (25% using 4-

iodonitrobenzene[24] and only 20% using iodobenzene[116] as the electrophile), together with a small amount of protonated material. A similar result was obtained with the glutamic acid derived homologue **111**. The poor mass balance was of particular concern and this prompted us to investigate these reagents further.

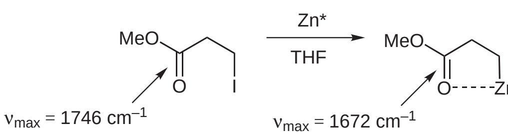
Inspection of ^{1}H NMR spectra of the reagents **110** and **111** in THF-d_8 revealed that significant β-elimination of the NHBoc group had occurred, yielding methyl but-3-enoate and methyl pent-4-enoate, respectively. However, reagent **5a** was found to be less prone to β-elimination, requiring extended heating at 50 °C over three days to decompose fully. Although the three compounds are all β-amido organozinc reagents, they appeared to exhibit markedly different stabilities.[116]

A comparison of the ^{13}C NMR chemical shifts of the ester carbonyl groups of the zinc reagent **5a** and its precursor iodide revealed downfield shifts upon zinc reagent formation, probably indicative of intramolecular coordination (Scheme 44).



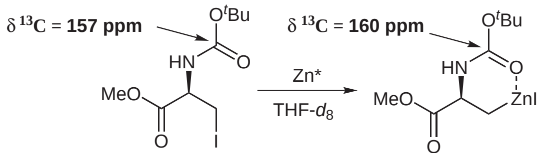
Scheme 44

This data is supported by changes in IR stretching frequencies upon forming a zinc reagent from 3-iodopropionic acid methyl ester, which showed a weakening of the carbonyl groups upon zinc insertion (Scheme 45),[117] consistent with literature precedent.[118]



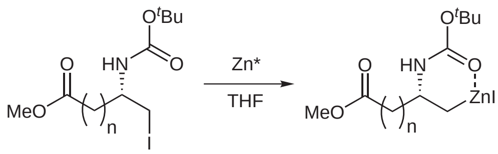
Scheme 45

In addition to coordination by the ester, the carbamate was also found to interact with zinc, although the interaction appeared to be less strong (Scheme 46).[116]



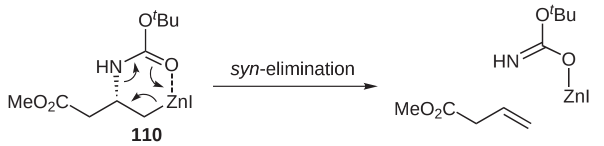
Scheme 46

Examination of the ^{13}C NMR spectra of the homologues revealed that internal coordination was also taking place, and the strength of these interactions was found to vary throughout the series. Thus, ester coordination was found to be especially strong with organozinc reagent **5a**, and was found to decrease through the series, as n changes from 0 to 2 and the ring size changes from 5 through to 7 (Scheme 47). Correspondingly, decreased ester coordination appears to lead to increased carbamate coordination (Table 1).[116]



Scheme 47

With the evidence so far, we reasoned that the decomposition of reagent **110** was occurring via a *syn*-elimination process, in which coordination of the zinc atom to the carbamate was important (Scheme 48).



Scheme 48

In contrast, it appeared that the strong ester coordination of zinc reagent **5a**, which appears to cause a reduction in zinc–carbamate interaction, was responsible for the relatively high stability of the reagent towards β-elimination. Thus, as carbamate coordination increases throughout the homologous series the elimination becomes more facile.

Table 1 Changes in ^{13}C NMR Chemical Shift of Carbonyl Groups upon Zinc Reagent Formation in THF-d_8

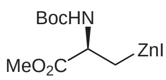
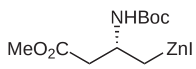
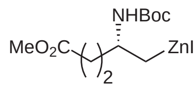
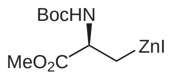
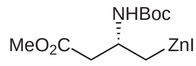
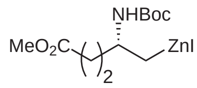
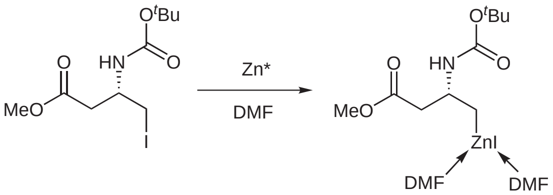
	Δδ [$\delta_{(R-ZnI)} - \delta_{(R-I)}$]  **5a**	 **110**	 **111**
Carbamate	+2.7	+3.7	+3.9
Ester	+5.3	+1.5	+0.7

Table 2 Changes in ^{13}C NMR Chemical Shift of Carbonyl Groups upon Zinc Reagent Formation in DMF-d_7

	$\Delta\delta\ [\delta_{(R-ZnI)} - \delta_{(R-I)}]$		
			
	5a	**110**	**111**
Carbamate	–0.9	–0.5	–0.7
Ester	+5.8	+0.9	+0.1

Preparation of the zinc reagents in DMF results in a decreased zinc–carbamate interaction, as measured by NMR spectroscopy. We ascribed this to a disruption of intramolecular coordination by the strongly Lewis basic solvent (Scheme 49), although a strong zinc–ester interaction was still observed for **5a** (Table 2).[116]



Scheme 49

As a consequence, much less eliminated material was observed for zinc reagents **110** and **111** when prepared in DMF than when they were prepared in THF. The decomposition of **110** in DMF was found to be approximately 4 times slower than in THF.[119]

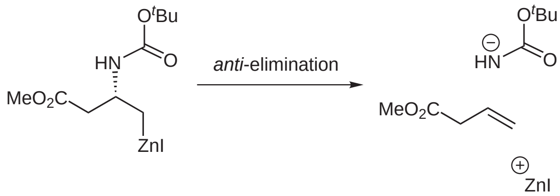
Since we had been able to record NMR spectra of the zinc reagents in solution, we decided to investigate whether the decomposition might be amenable to kinetic analysis. We found this elimination reaction to be a well-behaved first order process. Measurement of the rate constants at different temperatures in both solvents allowed the activation parameters for the elimination to be determined according to the Arrhenius equation (Table 3).[119]

Table 3 Activation Parameters for the Decomposition of Zinc Reagent **110** in THF and DMF

	Rate constant at 25 °C, k (s^{-1})	ΔH (J K^{-1} mol^{-1})	ΔS (kJ mol^{-1})
THF-d_8	1.02×10^{-4}	+70	–85
DMF-d_7	0.26×10^{-4}	+90	–32

The fact that ΔS is negative for a first-order reaction in both solvents suggests that the elimination is a *syn* process, since if it were *anti* we might expect an increase in entropy as there would be three developing species in the transition state (Scheme 50) rather than two (Scheme 48).

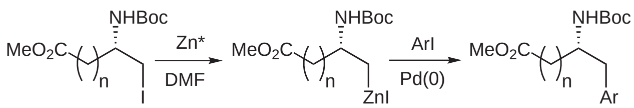
The smaller enthalpy of activation observed in THF is most easily explained by intramolecular coordination of



Scheme 50

the carbamate group to the zinc, apparently required for elimination, already being present to some extent in the ground state. In DMF such coordination in the ground state is apparently less significant (as already observed by $\Delta\delta$ in the ^{13}C NMR spectra), presumably due to intermolecular coordination of the zinc by solvent DMF. The need to lose this solvation (at least partially) before elimination can occur results in a higher enthalpy of activation in DMF. The most reasonable explanation for the observation that ΔS is more negative in THF is that while in the ground state there is strong intramolecular coordination between the carbamate and zinc, increased solvation of the zinc is required as the transition state is reached. In DMF, the zinc is already solvated in the ground state, and the decrease in entropy resulting from coordination of the carbamate to zinc can be partially offset by the increase of entropy due to concurrent loss of coordinated DMF as the transition state forms.[105,119]

Following this demonstration of the enhanced stability of these reagents in DMF, our group was then able to prepare a range of β-homophenylalanine and γ-bishomophenylalanine derivatives **112** and **113** in good yields by the Negishi reaction of zinc reagents **110** and **111** (n = 1, ca. 35–75%; n = 2, ca. 55–70%, Scheme 51).[24]



108, n = 1 **110**, n = 1 **112**, n = 1
109, n = 2 **111**, n = 2 **113**, n = 2

Scheme 51

Titration of a THF-d_8 solution of zinc reagent **5a** with DMF-d_7 showed a convergence of the diastereotopic protons, and with 4.0 equivalents of DMF-d_7 (relative to **5a**) the spectrum appeared similar to the spectrum in neat

DMF-d_7 (Figure 16). This demonstrated that the carbamate coordination could be significantly disrupted with relatively low concentrations of DMF and appreciable stabilisation towards β-elimination was to be expected.

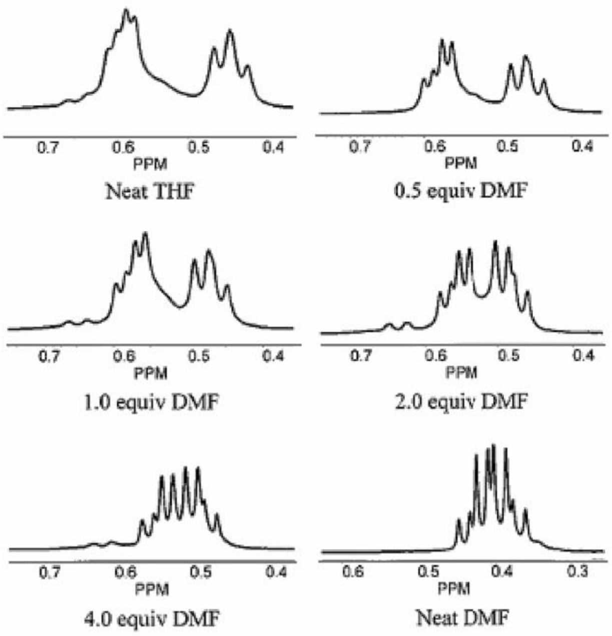


Figure 16 Convergence of the diastereotopic methylene protons upon addition of DMF-d_7 to a solution of zinc reagent **5a** in THF-d_8.

Our conclusion from these studies is that the serine-derived organozinc reagents **5** adopt an internally chelated structure in THF, with appreciable coordination of the zinc to both the carbamate and ester carbonyl oxygens. Although strong carbamate coordination might be expected to promote *syn* elimination, the strong ester carbonyl coordination ensures that the required orientation for elimination (either *syn* or *anti*) cannot be easily achieved. In DMF, while ester coordination evidently persists, the disruption of internal carbamate coordination appears to be beneficial to reactivity. In conclusion, the solution behaviour of these reagents is probably best described as a solvent dependent equilibrium (Figure 17). This is, of course, something of a simplification, since aggregation of alkylzinc halides is highly likely.

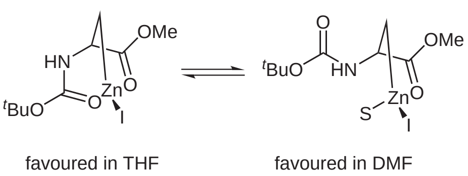
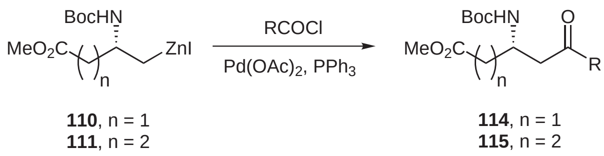


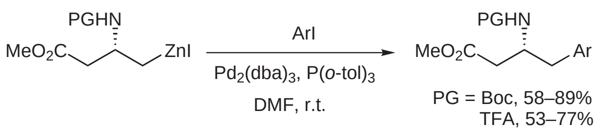
Figure 17

Following this discovery, and given that THF is unsuitable as a solvent for acid chloride cross-coupling reactions, our group decided to evaluate the acylation of zinc reagents **110** and **111** in a toluene–DMA mixture (2.0 equiv DMA).[119] The DMA was chosen since it has similar solvent properties to DMF, but is less reactive towards acid chlorides than DMF. Ketones **114** and **115** were each obtained in moderate (ca. 50%) yield (Scheme 52).



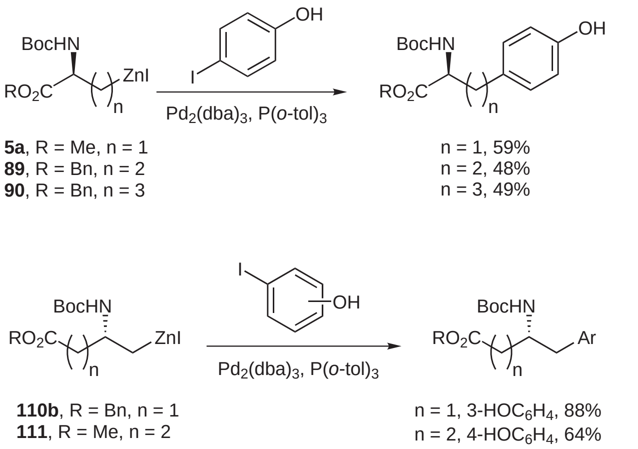
Scheme 52

During our investigations into the factors which influence the elimination of these organozinc reagents, we synthesised and reacted a range of amino acid derived organozinc reagents with the amine protected as a trifluoroacetamide (TFA) instead of the more usual Boc group.[120] A better leaving group, as indicated by a lower pK_a of the corresponding conjugate acid, would normally be expected to result in a faster rate of elimination. If, however, the elimination were dependent upon initial coordination between the zinc and the carbonyl of the leaving group, as we had proposed, the opposite effect would be observed. In the event, the results show that making the protected amine a 'better' leaving group actually results in stabilisation of the β-amido organozinc reagent towards elimination. In addition, there is no significant influence upon the yield of the cross-coupling reaction, compared to Boc protecting group (Scheme 53).[120] This provides good circumstantial evidence for our proposal that elimination is indeed a *syn* process.



Scheme 53

One additional consequence of the use of the TFA protecting group is that NH proton is significantly more acidic. The results shown in Scheme 53 led us to consider whether other protons of comparable acidity, such as in phenol, are tolerated in the cross-coupling reaction of these systems. A series of reactions has demonstrated that a range of unprotected iodophenols may be used successfully in the Pd-catalysed cross-coupling reaction of α-, β- and γ-amidozinc reagents derived from serine, aspartic acid and glutamic acid (Scheme 54).[121] In general, moderate to good yields were obtained with 3- and 4-iodophenol with various zinc reagents. Lower yields were obtained with 2-iodophenol, which simply reflects the behaviour of similar 2-substituted iodobenzene derivatives in these coupling reactions.[105]



Scheme 54

9 Conclusions

It has become clear during the course of this work that the functional group tolerance of the carbon–zinc bond is very high, probably more so than might have originally been expected. The corollary of this tolerance is that amino acid derived organozinc halides are intrinsically rather unreactive towards electrophiles, a drawback that has been overcome by transmetallation to palladium (in a catalytic sense) and copper (either in a stoichiometric or catalytic sense). On reflection, we were rather lucky to initiate our work using serine as starting material, since the iodoalanine-derived zinc reagent **5** appears to be amongst the most robust of the reagents we have prepared, although this robustness is reflected in the generally modest yields that can be obtained upon reaction with electrophiles. The other reagents that we have developed are more reactive, and provided the reagents are stabilised by the presence of suitable coordinating solvent, they can react efficiently with a variety of electrophiles. The ease of preparation of a range of organozinc iodides makes their application one of the more general approaches available to the synthesis of a wide range of non-natural amino acids.

Acknowledgment

We thank all the members of the group who have carried out the research described in this account, and also a number of industrial collaborators whose input has been invaluable, and all of whose names are included in the references. We are especially grateful to Dr. Anthony Wood (now at Pfizer Central Research, Sandwich), who first made the serine-derived zinc reagent, and Drs. Keith James and Martin Wythes also of Pfizer, for stimulating discussions and suggestions that were critical to the initial development of the project.

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